Utilizing Control Lyapunov Function and Control Barrier Function for Adaptive Cruise Control

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***Abstract*—This paper presents a reproducibility study of the Control Lyapunov Function (CLF) and Control Barrier Function (CBF) framework applied to the Adaptive Cruise Control (ACC) problem. The CLF encodes the performance objective, while the CBF enforces forward invariance of the safe set, and these are unified through a Quadratic Program (QP) that minimizes control effort subject to both constraints. Two cases are evaluated: one with only CLF and CBF constraints, and another incorporating force-based CBF (FCBF) and control constraints (CC). Simulation results demonstrate that without force constraints, the wheel force significantly exceeds comfort limits, whereas the inclusion of FCBF and CC ensures that the control input remains within acceptable bounds, thereby maintaining passenger comfort. Additionally, the effect of the relaxation penalty weight on the aggressiveness of speed tracking is analyzed.**

***Index Terms*—Adaptive Cruise Control, Control Lyapunov Function, Control Barrier Function, Quadratic Programming, Safety-Critical Control**

I. Introduction

Adaptive Cruise Control (ACC) is a common technology seen in modern cars. It is a safety-critical control problem that involves two inherently conflicting objectives: maintaining a desired cruising speed, which is the performance requirement, and preserving a safe following distance from the preceding vehicle, which is the safety requirement.

Prior approaches have addressed this problem through two primary frameworks. The first is the Proportional-Integral-Derivative (PID) controller, a classical control strategy in which the control input is computed based on the proportional, integral, and derivative terms of the tracking error [1]. While straightforward to implement, the PID framework does not explicitly account for safety constraints, making it insufficient for the ACC problem where collision avoidance is paramount. The second approach employs Model Predictive Control (MPC), which solves a receding horizon optimization problem at each time step [2]. In MPC, state constraints define the safe operating region by bounding the distance error, velocity error, and vehicle speed. However, these safety constraints are only enforced when the optimization problem remains feasible; if infeasibility arises, the safety guarantees are lost.

To address the limitations of both approaches, this work adopts the Control Lyapunov Function (CLF) and Control Barrier Function (CBF) framework proposed in [3]. The CLF encodes the performance objective by driving the system toward the desired speed, while the CBF enforces forward invariance of the safe set, thereby guaranteeing safety at all times. These two objectives are unified through a Quadratic Program (QP), which minimizes control effort subject to both stability and safety constraints.

II. Mathematical Foundation

A. Lyapunov Function

Lyapunov stability theory provides a fundamental tool for analyzing the stability of dynamical systems without explicitly solving the governing differential equations

Consider an autonomous nonlinear system as the form

$$\dot{x} = f(x) \tag{1}$$

where $f : R^n \to R^n$ is locally Lipschitz.

A Lyapunov function can be interpreted as a generalized energy function of the system. $V(x)$. It is a continuously differentiable, positive definite function satisfying the following conditions:

$$V(x) \tag{2}$$

$$\text{s.t. } V(x_e) = 0, \quad V(x) > 0 \text{ for } x \neq x_e, \tag{3}$$

$$\dot{V}(x) = \frac{\partial V}{\partial x} f(x) < 0 \text{ for } x \neq x_e \tag{4}$$

There conditions encode two important properties. First, the function is zero at the equilbrium point x_e and strictly positive elsewhere. This ensures that there is a valid measure of the system's deviation from equilibrium. Second, the time derivative $\dot{V}(x)$ is strictly negative, implying that the energy of the system monotonically decreases along system trajectories.

B. Barrier Function

While Lyapunov functions address stability, barrier function address safety by enforcing forward invariance. In other words, if the system state originates within the safe set, it remains within it for all future time.

Given a continuously differentiable function $h : \mathbb{R}^n \to \mathbb{R}$, where $h(x)$ quantifies the safety margin of the current state

x, the safe set $\mathcal{C}$ and its associated boundary and interior are defined as:

$$\mathcal{C} = \{x \in \mathbb{R}^n \mid h(x) \geq 0\} \tag{5}$$

$$\partial\mathcal{C} = \{x \in \mathbb{R}^n \mid h(x) = 0\} \tag{6}$$

$$\text{Int}(\mathcal{C}) = \{x \in \mathbb{R}^n \mid h(x) > 0\}. \tag{7}$$

A suitable barrier function candidate is given by:

$$B(x) = -\log\left(\frac{h(x)}{1+h(x)}\right) \tag{8}$$

A key property of this function is that as $h(x) \to 0^+$, $B(x) \to +\infty$. This creates an infinite penalty as the system state approaches the boundary of the safe set. To ensure forward invariance, the time derivative of $B(x)$ is requires to satisfy :

$$\dot{B}(x) \leq \frac{\gamma}{B(x)}, \tag{9}$$

Where γ is a parameter one can tune. This condition guarantees that as $B(x) \to \infty$ (i.e., the state approaches the unsafe boundary), $\dot{B}(x) \leq 0$, ensuring that the system is driven away from the boundary and remains within the safe set $\mathcal{C}$.

C. Control Barrier Function

While the barrier function introduced in the previous section ensures safety for autonomous systems, it doesn't account for the presence of control inputs. The Control Barrier Function (CBF) extends this concept to control-affine systems, enabling the controller that guarantee forward invariance.

Consider a control-affine nonlinear system of the form

$$\dot{x} = f(x) + g(x)u \tag{10}$$

With f and g locally Lipschitz, $x \in \mathbb{R}^m$ denotes the state, and $u \in \mathbb{R}^m$ denote the control input. The goal is to design a controller to ensure containment in $\mathcal{C}$

Let $\mathcal{C} \subset \mathbb{R}^n$ be defined by (5)–(7) for a continuously differentiable function $h : \mathbb{R}^n \to \mathbb{R}$. A function $B : \text{Int}(\mathcal{C}) \to \mathbb{R}$ is a **control barrier function (CBF)** if there exist class $\mathcal{K}$ functions α_1, α_2 and a constant $\gamma > 0$ such that, for all $x \in \text{Int}(\mathcal{C})$:

$$\frac{1}{\alpha_1(h(x))} \leq B(x) \leq \frac{1}{\alpha_2(h(x))}, \tag{11}$$

$$\inf_{u \in U}\left[L_f B(x) + L_g B(x)\, u - \frac{\gamma}{B(x)}\right] \leq 0, \tag{12}$$

where $L_f B(x)$ and $L_g B(x)$ denote the Lie derivatives of $B(x)$ along f and g, respectively. The Lie derivative captures how $B(x)$ evolves along the system dynamics: $L_f B(x)$ represents the effect of the drift dynamics, while $L_g B(x)$ captures the influence of the control input. The derivation $\dot{B}(x)$ is included in supplementary material (Section A).

Condition (11) ensures that $B(x)$ is bounded between two inverse of class-κ functions that grow to infinity as the state approaches the boundary of $\mathcal{C}$. Condition (12) guarantees the existence of a control input u that prevents the state from reaching the boundary, thereby enforcing forward invariance of $\mathcal{C}$.

D. CLF and ES-CLF

Analogous to how the CBF extends the barrier function to incorporate control inputs, the Control Lyapunov Function (CLF) extends the classical Lyapunov function to controlled systems by ensuring the existence of a control input that drives the system toward equilibrium.

Let $V(x) : \mathbb{R}^n \to \mathbb{R}$ be a continuously differentiable function. $V(x)$ is a (local) Control Lyapunov Function (CLF) if there exists a constant $c > 0$ such that the following conditions hold:

1) **Bounded sublevel set**

$$\Omega_c := \{x \in \mathbb{R}^n : V(x) \leq c\}$$

2) **Positive definiteness**

$$V(x) > 0 \text{ for all } x \neq x_e, \qquad V(x_e) = 0$$

3) **Control-dependent decrease condition**

$$\min_{u \in U} \dot{V}(x,u) < 0 \quad \forall x \in \Omega_c \setminus \{x_e\}$$

The CLF ensures that there exists a control input u to make $V(x)$ decrease over time, thereby ensuring asymptotic stability. However, it doesn't guarantee the rate of convergence. To address this limitation, Exponentially Stabilizing Control Lyapunov Function (ES-CLF) further imposes a stronger condition to ensure an exponential rate of decrease. Specifically, V(x) is **Exponentially Stabilizing-Control Lyapunov Function (ES-CLF)** if there exists a constant $\lambda > 0$ such that

$$\min_{u \in U} \dot{V}(x,u) + \lambda V(x) \leq 0. \tag{13}$$

After solving the differential equation, one can see that $V(x)$ decreases at least exponentially, i.e., $V(x(t)) \leq V(x(0))\, e^{-\lambda t}$, providing a quantifiable bound on how fast the system converges to equilibrium.

E. CLF-CBF QP

The CLF and CBF formulations presented in the preceding subsections encode performance and safety objectives independently. To construct a controller that achieve both objectives, quadratic program (QP) can be utilized to unify the constraints. The QP minimizes control effort while enforcing stability through the CLF constraint and safety through the CBF constraint. The resulting optimization problem is formulated as:

$$u^*(x,z) = \underset{\mathbf{u}=\begin{bmatrix} u \\ \delta \end{bmatrix} \in \mathbb{R}^{m+1}}{argmin} \quad \frac{1}{2}\mathbf{u}^T H(x,z)\mathbf{u} + F(x,z)^T\mathbf{u} \tag{CLF-CBF QP}$$

$$\text{s.t.} \quad \psi_0(x,z) + \psi_1^T(x,z)u \leq \delta \tag{CLF}$$

$$L_f B(x,z) + L_g B(x,z)u \leq \frac{\gamma}{B(x,z)} \tag{CBF}$$

where $H(x,z) \succ 0$ is a positive definite weight matrix, $F(x,z)$ is a linear cost vector, and δ is a relaxation variable that allows the CLF constraint to be softened when it conflicts with the safety-critical CBF constraint.

III. Math Formulation

A. Problem Setup

The scenario consists of two vehicles: an ego vehicle controlled by the ACC system, and a leading vehicle traveling at a constant velocity v_0. For the ego vehicle, let $x = (x_1, x_2)$, where x_1 denotes the longitudinal position and $x_2 = \dot{x}_1$ denotes the velocity. Define $z = D$ as the inter-vehicle distance. The dynamics of the system are modeled as a control-affine system

$$\dot{x} = \underbrace{\begin{bmatrix} x_2 \\ -\frac{1}{m}F_r \end{bmatrix}}_{f(x,z)} + \underbrace{\begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix}}_{g(x,z)} u \tag{14}$$

$$\dot{z} = \underbrace{v_0 - x_2}_{q(x,z)}, \tag{15}$$

In (14), the first row indicates that position evolves according to the vehicle's velocity, while the second row captures Newton's second law: acceleration is determined by the net force (control input minus resistance) divided by the vehicle mass. Equation (15) describes the evolution of the inter-vehicle distance, which decreases when the ego vehicle travels faster than the leading vehicle.

The first row of the constraint means the position changes with velocity of the ego car Position changes according to velocity, and the second row is the acceleration is related to the proportion of the control force minus the resistance. The distance changes based on the speed difference between the front vehicle and the host vehicle.

B. Constraint

Four types of constraints are imposed on the QP to encode safety, performance, comfort, and physical limitations of the vehicle.

Hard Constraint as CBF. The primary safety objective is to maintain a minimum following distance from the leading vehicle. Following the half-the-speedometer rule, the safe distance requirement is expressed as:

$$D \geq \frac{v}{2} \tag{16}$$

This safety requirement is encoded as a CBF constraint of the form:

$$L_f B(x,z) + L_g B(x,z)\, u - \frac{\gamma}{B(x,z)} \leq 0, \qquad \text{(HC1-CBF)}$$

where the barrier function and its associated Lie derivatives are defined as:

$$h(x,z) = z - 1.8x_2 \tag{17}$$

$$B(x,z) = -log(\frac{h(x)}{1+h(x)}) \tag{18}$$

$$L_f B(x,z) = -\frac{1.8F_r(x) + m(v_0 - x_2)}{m(1 - 1.8x_2 + z)(-1.8x_2 + z)} \tag{19}$$

$$L_g B(x,z) = \frac{1.8}{m(1 - 1.8x_2 + z)(-1.8x_2 + z)} \tag{20}$$

The derivations of the lie derivative form are provided in the supplementary material (Section B).

Soft Constraint as CLF. The performance objective is to drive the ego vehicle's velocity toward the desired cruising speed v_d:

$$v - v_d \to 0 \tag{21}$$

This objective is encoded as a CLF constraint derived from the ES-CLF condition (cf. Equation 13):

$$\psi_0(x,z) + \psi_1(x,z)\, u \leq \delta_{sc}, \qquad \text{(SC1-CLF)}$$

where δ_{sc} is a relaxation variable that allows the CLF constraint to be softened when it conflicts with the safety-critical CBF constraint, and

$$\psi_0(x,z) = -\frac{2(x_2 - v_d)}{m} F_r(x) + \varepsilon(x_2 - v_d)^2 \tag{22}$$

$$\psi_1(x,z) = \frac{2(x_2 - v_d)}{m} \tag{23}$$

The complete derivations of SC1-CLF are provided in the supplementary material (Section C).

Force-based constraint as CBF. In practice, the wheel forces applied by an ACC system should remain well below the vehicle's peak force capacity. The reason is that ACC is not emergency control. Excessive control forces lead to abrupt acceleration or braking, causing passenger discomfort and decreasing driver trust in the system.

To enforce comfort limits, a force-based CBF (FCBF) constraint is imposed:

$$L_f B_F(x,z) + L_g B_F(x,z)\, u - \frac{1}{B_F(x,z)} \leq 0, \qquad \text{(FCBF)}$$

where:

$$B_F(x,z) = \frac{1}{h_F(x,z)}, \tag{24}$$

$$h_F(x,z) = z - 1.8\, x_2 - \frac{(v_0 - x_2)^2}{2\, c_d g}, \tag{25}$$

$$L_f B_F = -\frac{1}{h_F(x,z)^2} \left[\left(1.8 - \frac{v_0 - x_2}{c_d g}\right) \frac{F_r}{m} + (v_0 - x_2) \right], \tag{26}$$

$$L_g B_F = \frac{-1}{m\, h_F^2} \left(-1.8 + \frac{v_0 - x_2}{c_d g} \right). \tag{27}$$

The function $h_F(x,z)$ in (25) incorporates both the required safety gap ($1.8\, x_2$) and the braking distance under constant deceleration $c_d g$, providing a more conservative safety margin

than $h(x, z)$ in (17). The complete derivations are provided in the supplementary material (Section D).

Control Constraint. The physical actuator limits of the vehicle bound the maximum allowable acceleration and deceleration forces:

$$u \leq c_a mg, \tag{FC1}$$
$$-u \leq c_d mg, \tag{FC2}$$

where c_a and c_d are dimensionless coefficients representing the fraction of gravitational force available for acceleration and braking, respectively, and g is the gravitational constant. The proof is listed in E.

C. Overall Formulation

Combining all four constraint types, the complete CLF-CBF QP for adaptive cruise control is formulated as:

$$\begin{aligned} u^*(x,z) = \quad & \underset{\mathbf{u}=\begin{bmatrix} u \\ \delta_{sc} \end{bmatrix} \in \mathbb{R}^2}{\arg\min} && \frac{1}{2}\mathbf{u}^T H_{\text{acc}} \mathbf{u} + F_{\text{acc}}^T \mathbf{u} \\ & \text{s.t.} && A_{\text{clf}}\mathbf{u} \leq b_{\text{clf}} && \text{(CLF)} \\ & && A_{\text{cbf}}\mathbf{u} \leq b_{\text{cbf}} && \text{(CBF)} \\ & && A_{\text{fcbf}}\mathbf{u} \leq b_{\text{fcbf}} && \text{(FCBF)} \\ & && A_{\text{fc}}\mathbf{u} \leq b_{\text{fc}} && \text{(CC)} \end{aligned}$$

where the constraint matrices and vectors are defined as follows:

$$\begin{aligned} A_{\text{clf}} &= \begin{bmatrix} \psi_1(x,z) & -1 \end{bmatrix}, \\ b_{\text{clf}} &= -\psi_0(x,z). \end{aligned}$$

$$\begin{aligned} A_{\text{cbf}} &= \begin{bmatrix} L_g B(x,z) & 0 \end{bmatrix}, \\ b_{\text{cbf}} &= -L_f B(x,z) + \frac{\gamma}{B(x,z)}. \end{aligned}$$

$$A_{\text{cc}} = \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}, \qquad b_{\text{cc}} = \begin{bmatrix} c_a mg \\ c_d mg \end{bmatrix}.$$

$$\begin{aligned} A_{\text{fcbf}} &= \begin{bmatrix} L_g B_F(x,z) & 0 \end{bmatrix}, \\ b_{\text{fcbf}} &= -L_f B_F(x,z) + \frac{1}{B_F(x,z)}. \end{aligned}$$

IV. Result

The proposed CLF-CBF QP controller is evaluated through numerical simulations using the parameters listed in Table I. The simulation environment used is Google Colab, utilizing the qpsolver, math, and numpy Python packages. Two cases are evaluated to demonstrate the effect of incorporating force-based constraints, and the influence of the relaxation penalty weight p_{sc} on system behavior is analyzed.

- **Case I:** The QP is solved subject to only the CLF and CBF constraints. The CBF enforces forward invariance based on the half-the-speedometer rule (cf. (16)), while the CLF drives the ego vehicle toward the desired speed v_d when the safety constraint is satisfied.
- **Case II:** In addition to the Case I constraints, the force-based CBF (FCBF) and control constraints (CC) are included in the QP formulation, bounding the wheel force within comfort limits.

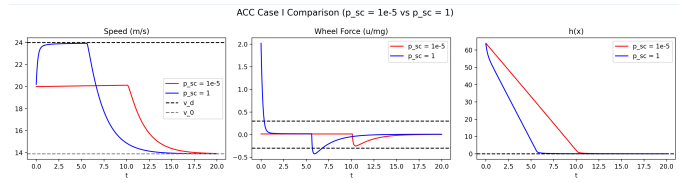

Fig. 1. Case I simulation result

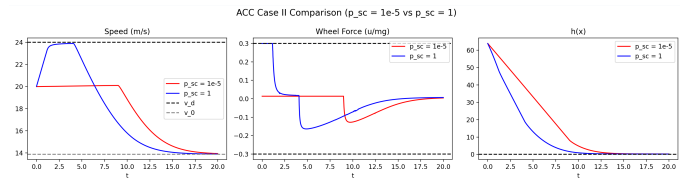

Fig. 2. Case II simulation result

To understand the role of p_{sc}, consider the expanded cost function. Substituting the structure of H_{acc} into the QP objective yields:

$$\frac{1}{2}\mathbf{u}^T H_{\text{acc}} \mathbf{u} = \frac{1}{2}\begin{bmatrix} u & \delta_{sc} \end{bmatrix} \cdot 2 \begin{bmatrix} \frac{1}{m^2} & 0 \\ 0 & p_{sc} \end{bmatrix} \cdot \begin{bmatrix} u \\ \delta_{sc} \end{bmatrix} = \frac{u^2}{m^2} + p_{sc}\delta_{sc}^2. \tag{28}$$

Since δ_{sc} is the relaxation variable associated with the ES-CLF constraint (cf. (29)):

$$\dot{V} + \varepsilon V \leq \delta_{sc}, \tag{29}$$

the weight p_{sc} directly controls how aggressively the system penalizes CLF relaxation.

Fig. 1 presents the simulation results for Case I. When $p_{sc} = 1$, the controller aggressively increases the ego vehicle's speed toward $v_d = 24$ m/s. However, as shown in the second subfigure, the resulting wheel force significantly exceeds the comfort constraint (indicated by the dotted line), which would cause abrupt acceleration and braking that is unacceptable for passenger comfort. Fig. 2 presents the simulation results for Case II, which includes the FCBF and CC constraints. In contrast to Case I, the wheel force remains within the prescribed comfort bounds at all times, demonstrating the effectiveness of the force-based constraints in maintaining passenger comfort

Despite the difference in constraint sets, both cases exhibit a similar qualitative trend. As shown in Figs. 1 and 2, the ego vehicle initially increases its speed toward v_d. However, as the inter-vehicle distance decreases and approaches the safety boundary defined by the CBF, the controller reduces the ego vehicle's speed to match that of the leading vehicle ($v_0 = 13.89$ m/s), thereby satisfying the hard safety constraint. This behavior confirms that the CBF successfully enforces forward

invariance of the safe set, preventing collision regardless of the CLF's performance objective.

V. Implementation Challenges

The original paper [3] presents the force-based CBF constraint solely in Lie derivative form (i.e., $L_f B_F$ and $L_g B_F$), without providing the explicit closed-form expressions. Consequently, the full analytical expressions for $L_f B_F$ (cf. (26)) and $L_g B_F$ (cf. (27)) had to be independently derived from the definition of $h_F(x, z)$ in (25) using the chain rule and the structure of the control-affine dynamics in (14). The complete derivations are provided in the supplementary material (Section D).

Two notable differences exist between the reproduction and the original work. First, the original paper only presents simulation results for $p_{sc} = 10^{-5}$, whereas this work additionally evaluates the case $p_{sc} = 1$ to illustrate the effect of aggressive speed tracking on wheel force behavior. Second, while the reproduced figures do not exactly match those in the original paper, the qualitative trends, including the speed convergence behavior and the effect of the CBF on maintaining safe following distance, are consistent with the reported results.

VI. Conclusion

The paper presented a reproducibility study of the CLF-CBF QP framework apoplied to the Adaptive Cruise Control problem. By including the full constraints (CLF, CBF, FCBF, CC), the controller is able to perform braking without causing discomfort. Furthermore, the analysis of the relaxation penalty weight p_{sc} revealed its direct influence on the aggressiveness of speed tracking: larger values of p_{sc} penalize CLF relaxation more heavily, compelling the system to pursue the desired speed more aggressively.

Several directions for future investigation are identified:

1) **Varying lead-vehicle speed:** The current simulation assumes the leading vehicle travels at a constant velocity $v_0 = 13.89$ m/s [2]. Extending the framework to accommodate time-varying lead-vehicle speeds would provide a more realistic evaluation of the controller's performance.
2) **Sensor noise:** The current implementation assumes perfect state measurements. In practice, sensor measurements are subject to noise, which can degrade controller performance. Incorporating state estimation techniques such as Kalman filters [4] would improve the robustness and practical applicability of the approach.
3) **Baseline comparisons:** A rigorous quantitative comparison against PID [1] and MPC [2] baselines, like using metrics such as tracking error, constraint violation frequency, etc, would provide a more comprehensive assessment of the CLF-CBF QP framework's advantages.

VII. Code and Data Availability

The implementation of the module and the resulting visualization can be seen in the following repository. https://github.com/Frankkao813/AIRobotics-FP-CLF_CBF_ACC

Parameter	Value	Units
g	9.81	$\frac{m}{s^2}$
m	1650	kg
f_0	0.1	$Newton$
f_1	5	$\frac{(Newton)(s)}{m}$
f_2	0.25	$\frac{(Newton)(s^2)}{m}$
v_d	24	$\frac{m}{s}$
v_0	13.89	$\frac{m}{s}$
ε	10	Unitless
γ	1	Unitless
c_a	0.3	Unitless
c_d	0.3	Unitless
p_{sc}	$1e^{-5}$	Unitless

TABLE I
Table of parameters used in the simulation results.

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Appendix

In the following proof, the augmented state is constructed as $\bar{x} = (x_1, x_2, z)$, and then

$$\dot{\bar{x}} = \begin{bmatrix} \dot{x_1} \\ \dot{x_2} \\ \dot{z} \end{bmatrix} = \underbrace{\begin{bmatrix} x_2 \\ \frac{-1}{m} F_r \\ v_0 - x_2 \end{bmatrix}}_{f(x,z)} + \underbrace{\begin{bmatrix} 0 \\ \frac{1}{m} \\ 0 \end{bmatrix}}_{g(x,z)} u \tag{30}$$

A. Lie Derivatives and Control-Affine Systems

Suppose we have a affine control system

$$\dot{x} = f(x) + g(x)u \tag{31}$$

and a barrier function

$$B(x) \tag{32}$$

We want to know that how does $B(x)$ change over time as the state evolves.

$$\dot{B}(x) = \nabla B(x)\dot{x} = \nabla B(x) \cdot (f(x) + g(x)u) \tag{33}$$

We define the following, which is called the lie derivatives.

$$L_f B(x) := \nabla B(x) \cdot f(x) \tag{34}$$

$$L_g B(x) := \nabla B(x) \cdot g(x) \tag{35}$$

Therefore, we have

$$\dot{B}(x) = L_f B(x) + L_g B(x) u \tag{36}$$

Intuitively, $L_f B(x)$ is the effect of the system dynamics, and $L_g B(x)$ is the effect of the control input

B. Constraint I proof

By the Lie Derivative, one have

$$\dot{B}(x,z) = L_f B(x,z) + L_g B(x,z) u \tag{37}$$

We also impose the constraint on $\dot{B}$ to be

$$\dot{B} \leq \frac{\gamma}{B} \tag{38}$$

Substituting and rearrangement to get

$$L_f B(x,z) + L_g B(x,z) u - \frac{\gamma}{B(x,z)} \leq 0 \tag{39}$$

C. Constraint II proof

Recall that the tracking error is $y = x_2 - v_d$. Define a Lyapunov function [5] as

$$V = y^2 = (x_2 - v_d)^2 \tag{40}$$

Differentiating and rearrangements yield

$$\dot{V} = 2y\dot{y} = 2(x_2 - v_d)(-\frac{1}{m}F_r(x) + \frac{1}{m}u) \tag{41}$$

$$\dot{V} = \frac{-2(x_2 - v_d)}{m}F_r(x) + \frac{2(x_2 - v_d)}{m}u \tag{42}$$

Inserting the relaxed ES-CLF condition $\dot{V} + \epsilon V \leq \delta_{sc}$

$$\frac{-2(x_2 - v_d)}{m}F_r(x) + \frac{2(x_2 - v_d)}{m}u + \epsilon(x_2 - v_d)^2 \leq \delta_{sc} \tag{43}$$

After rearrangement, we can get the expression of

$$\frac{-2(x_2 - v_d)}{m}F_r(x) + \frac{2(x_2 - v_d)}{m}u + \epsilon(x_2 - v_d)^2 \leq \delta_{sc} \tag{44}$$

where

$$\psi_0(x,z) = -\frac{2(x_2 - v_d)}{m}F_r(x) + \varepsilon(x_2 - v_d)^2 \tag{45}$$

$$\psi_1(x,z) = \frac{2(x_2 - v_d)}{m} \tag{46}$$

D. Constraint III proof

1) Derivation of h_F:

$$h_F(x,z) = \underbrace{z}_{\text{distance}} - \underbrace{1.8x_2}_{\text{required safety gap}} - \underbrace{\frac{1}{2}\frac{(v_0 - x_2)^2}{c_d g}}_{\text{braking distance}} \tag{47}$$

For the required safety gap term, The original formulation is $z \geq 1.8x_2$ We already know that

$$v_{m/s} = \frac{v_{km/h}}{3.6} \tag{48}$$

$$v_{km/h} = 3.6 v_{m/s} \tag{49}$$

$$\tag{50}$$

Since $D \geq \frac{v_{m/2}}{2}$, we have

$$D \geq 1.8 v_{m/s} \tag{51}$$

For the braking distance term, note that

$$d = \frac{v^2}{2a} \tag{52}$$

plug in the constant deaccerlation term $a = -c_d g$ and the relative speed $v_0 - x_2$, we can get the distance lost as

$$d = \frac{v_0 - x_2}{2c_d g} \tag{53}$$

2) $L_f B_F$: Since $B_F(x,z) = \frac{1}{h_F(x,z)}$,

$$\begin{aligned} L_f B_F &= \nabla B_F \cdot f \\ &= \frac{\partial B_F}{\partial h_F} \nabla h_F \cdot f \\ &= \left(-\frac{1}{h_F^2}\right) L_f h_F \end{aligned} \tag{54}$$

We now compute $L_f h_F$,

$$h_F(x,z) = z - 1.8x_2 - \frac{(v_0 - x_2)^2}{2c_d g} \tag{55}$$

$$\begin{aligned} \frac{\partial h_F}{\partial x_1} &= 0 \\ \frac{\partial h_F}{\partial x_2} &= -1.8 + \frac{v_0 - x_2}{c_d g} \\ \frac{\partial h_F}{\partial z} &= 1 \end{aligned} \tag{56}$$

$$\begin{aligned} L_f h_F &= \nabla h_F \cdot f \\ &= \begin{bmatrix} 0 \\ -1.8 + \frac{v_0 - x_2}{c_d g} \\ 1 \end{bmatrix} \cdot \begin{bmatrix} x_2 \\ -\frac{1}{m}F_r \\ v_0 - x_2 \end{bmatrix} \\ &= 0 \cdot x_2 + \left(-1.8 + \frac{v_0 - x_2}{c_d g}\right)\left(-\frac{1}{m}F_r\right) + 1 \cdot (v_0 - x_2) \\ &= \left(1.8 - \frac{v_0 - x_2}{c_d g}\right)\frac{F_r}{m} + (v_0 - x_2) \end{aligned} \tag{57}$$

We compute $L_f B_F$ together.

$$L_f B_F = -\frac{1}{h_F(x,z)^2}\left[\left(1.8 - \frac{v_0 - x_2}{c_d g}\right)\frac{F_r}{m} + (v_0 - x_2)\right] \tag{58}$$

3) $L_g B_F$:

$$\begin{aligned} L_g B_F &= \nabla B_F \cdot g \\ &= \frac{\partial B_F}{\partial h_F} \nabla h_F \cdot g \\ &= \left(-\frac{1}{h_F^2}\right) L_g h_F \end{aligned} \tag{59}$$

We now compute $L_g h_F$. Using the expression of h_F from (47), and the partial derivative result from (56)

$$\begin{aligned} L_f h_F &= \nabla h_F \cdot g \\ &= \begin{bmatrix} 0 \\ -1.8 + \frac{v_0 - x_2}{c_d g} \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ \frac{1}{m} \\ 0 \end{bmatrix} \\ &= \frac{1}{m}(-1.8 + \frac{v_0 - x_2}{c_d g}) \end{aligned} \tag{60}$$

We compute $L_g B_F$ together from (59) and (60)

$$L_g B_F = \frac{-1}{m h_F^2}(-1.8 + \frac{v_0 - x_2}{c_d g}) \tag{61}$$

E. Control Constraints

This section serves two purposes: first, to derive the allowable wheel force u from the physical limits given in (FC1) and (FC2); and second, to construct the constraint matrix A_{fc} and vector b_{fc} such that $A_{\text{fc}}\, u \leq b_{\text{fc}}$. By rearranging Newton's second law of motion, we have (62):

$$\begin{aligned} -c_d g \leq \frac{u}{m} \leq c_a g \\ -c_d m g \leq u \leq c_a m g \end{aligned} \tag{62}$$

$$\begin{cases} u \leq c_a m g \\ -u \leq c_d m g \end{cases} \tag{63}$$

$$\underbrace{\begin{bmatrix} 1 \\ -1 \end{bmatrix}}_{A_{\text{fc}}} u \;\leq\; \underbrace{\begin{bmatrix} c_a m g \\ c_d m g \end{bmatrix}}_{b_{\text{fc}}} \tag{64}$$